SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

Figure placeholder: concept figure, design pending

Planned content: one Dynamic-14 frame overlaid with its real training labels; no generated imagery. The target at capture (solid) and when the commands take effect (translucent); the target's SE(3) poses at the 8 forecast horizons (axes); sparse 3D flow on the 14×14 patch points (arrows, near zero on the background). Task, layout, and styling to be decided.

Figure 1. **Concept figure (placeholder).** The caption will be written with the final figure.

Abstract

Manipulating moving objects requires actions that fit the scene when they execute, not when the image was captured, yet every millisecond spent inferring the scene's motion lets the target move further. World–action models forecast future images or latents at uniform fidelity, spending time on appearance while leaving the target's metric pose implicit; target-centric predictors are fast but ignore the surroundings that constrain or drive the motion. We observe that the parts of a dynamic scene need different precision: the target's pose decides contact to within millimeters, the bodies around it matter only through the clearance they leave and the motion they impart to the target, and the remaining background does not matter. We present SE(3)-WAM, a world–action model built on this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history, so one RGB-D frame suffices as input. It learns to forecast the target's future SE(3) trajectory precisely and the whole scene coarsely as sparse 3D flow, with forecasts used only in training. Together, these choices let SE(3)-WAM act in [XX] ms on one RTX 5090, [N]× faster than video world–action baselines. We also introduce Dynamic-14, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference. SE(3)-WAM achieves [XX.X]% success on Dynamic-14, [XX.X] points above the strongest baseline, and [XX]% across six real-robot tasks with diverse target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [5, 6, 11, 19, 38, 44, 58], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. In such scenes, an action must fit the scene at the moment it executes, not the moment its observation was captured, so a policy has to know how the scene moves. That knowledge must itself be inferred, and inference takes time during which the scene keeps moving: the longer a policy computes to anticipate the motion, the further ahead it must anticipate.

Recent dynamic policies add a memory of past motion, typically image-space cues such as optical flow [13, 25], and a forecast of the future [25, 34, 51]. Video-based WAMs forecast whole future frames or latents at uniform fidelity, spending much of their computation on visual detail [38, 44]. Compact future codes [26] reduce this cost and metric 3D scene dynamics [49] add geometry, but both still treat the whole scene uniformly. In each case, the quantity that decides contact, the target's metric pose at execution time, stays implicit. Target-centric predictors and dynamic VLAs with object-level forecasts sit at the other end [13, 16]: they are fast but anticipate only the target, leaving out the surroundings that can block or redirect its motion. Neither approach allocates precision according to what contact requires from each part of the scene. This raises a central question: what must a policy know about a moving scene, and how precisely, to make contact while keeping inference latency low?

Our key observation is that a dynamic policy needs to know very little precisely (Fig. 1). Contact demands millimeter precision in the target's pose at execution time; the surroundings matter only through the clearance they leave and the motion they impart to the target, so they need only be known coarsely. Building on this observation, we introduce SE(3)-WAM, a world–action model that learns a role-aware metric 4D representation, precise where contact demands it (Fig. 2). To supply motion information without encod-

ing past image frames, SE(3)-WAM pairs a single RGB-D frame with a short history of the target's metric SE(3) poses. A world stream learns the representation by forecasting the target's future SE(3) trajectory and the motion of the whole scene as a sparse 3D flow. An action stream reads the world stream's scene and summary tokens to generate action chunks, with explicit forecasting confined to training [48]. SE(3)-WAM thus combines precise anticipation of target motion with low inference latency, limiting how much the scene changes between observation and action.

We build Dynamic-14, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference, so every method pays for its own latency. SE(3)-WAM achieves [XX.X]% success on Dynamic-14, [XX.X] points above the strongest baseline, at a p95 latency of [XX] ms, [N]× lower than that of video world–action baselines. It also leads on 24 DOMINO tasks [13] and succeeds in [XX]% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Controlled ablations test the contributions of target-motion history and future supervision, and compare role-aware with uniform-fidelity forecasts to examine the allocation of precision.

Our contributions are threefold:

- We show that a dynamic policy needs millimeter precision only in the target's pose at execution time, and instantiate this principle as role-aware 4D modeling: a precise metric SE(3) trajectory for the target and a coarse sparse 3D flow for the whole scene.
- We develop SE(3)-WAM, a world–action model that combines metric target-pose histories with role-aware predictive supervision for low-latency action generation.
- We introduce Dynamic-14, an open-source dynamic manipulation benchmark that charges each policy its own inference latency, and demonstrate gains over strong baselines on Dynamic-14, 24 DOMINO tasks, and six real-robot tasks.

2. Related Work

Dynamic manipulation. Compensating for the time between sensing and acting is a classical problem, from control with dead time [31] to catching systems that track the target with Kalman-type filters and plan an interception [1–3, 28]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes [42]. PUMA reads historical optical flow and is supervised with future object features [13], and DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [25]. Both infer past motion from image-space cues, whereas SE(3)-WAM takes the target's metric SE(3) pose history as input and observes only the current RGB-D frame. Other dynamic policies forecast salient tokens [34], future images [52], compact future codes [26], or 3D scene dynamics [49] and condition their actions on these forecasts at test time. DynaWM adapts a base VLA's action chunks for dynamic manipulation [10], and DynamicManip synthesizes dynamic demonstrations from a static one [21]. Execution schemes reduce or hide latency by overlapping inference with execution [30, 42], keeping consecutive chunks consistent [7, 23], or conditioning on the execution time [35]. They decide when actions are computed and applied, not what the policy represents. DOMINO [13] and PhysMani-Bench [49] provide diverse dynamic-manipulation tasks, but their released default evaluation pipelines advance the simulation only after a policy has returned its prediction. Dynamic-14 advances the scene during inference and charges each method its measured latency.

Predictive supervision in WAMs. World–action models (WAMs) generate future video and actions in one network [20, 38, 44, 56], building on video prediction as a policy objective [36, 41]. Because generating video is costly, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [51], flow and geometry alongside RGB [47], object masks or slots [24, 45], or compact latents [33, 55], and asks whether the forecast is needed at test time. Fast-WAM retains most of the benefit of video modeling without test-time imagination [48], and PUMA and SG-WAM drop their predictors after training [13, 53]. SE(3)-WAM keeps this training-only use but forecasts primarily a metric 4D state rather than pixels, features, or latents, so the target's future pose is supervised explicitly instead of being left implicit.

Object-centric and geometric motion representations. Metric 3D inputs make policies more precise and sample-efficient [29, 50]. Motion has been represented by 2D point tracks [4, 40, 43], 3D flow [14, 17, 46, 49], and object keypoints or SE(3) trajectories [16, 27, 32], while rigid-body decompositions have long been used to learn scene dynamics [8, 9]. These works describe the scene with one kind of representation, such as dense flow or tracks everywhere, or a pose for each object. Target-centric predictors, including dynamic VLAs with object-level forecasts, model only the target's future [13, 16] and leave out the bodies that block or push it. DOCIR separates the robot, the target, and obstacles, but uses these distinctions only as input encodings [12]. SE(3)-WAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, while the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast.

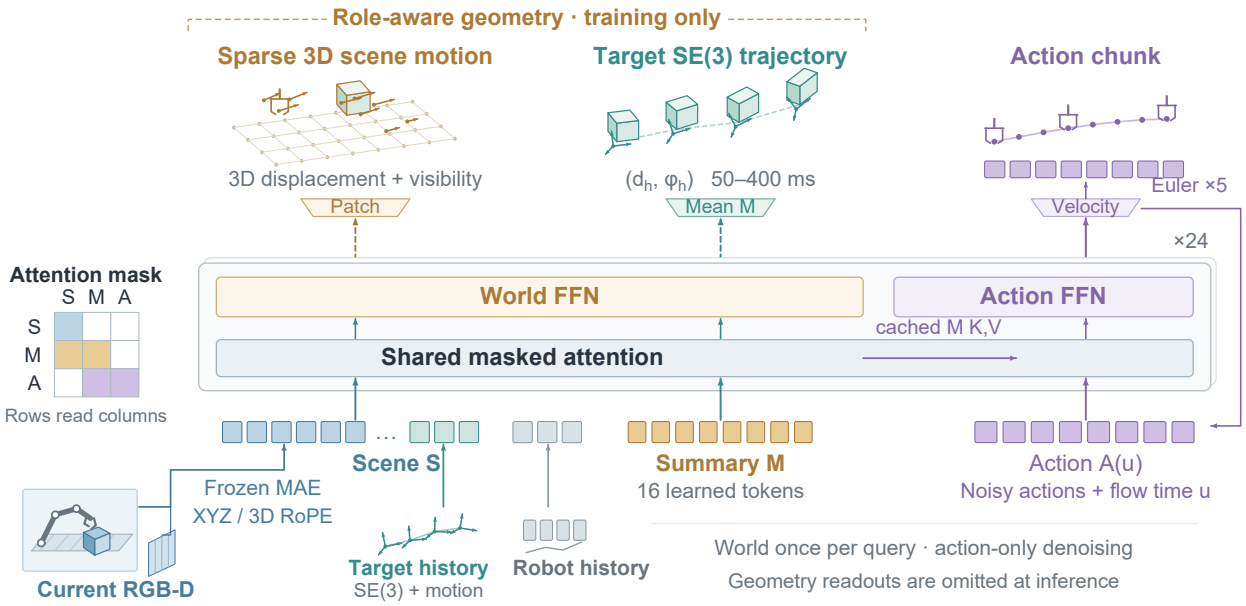


Figure 2. **SE(3)-WAM architecture.** Read from bottom to top. Current RGB-D and target/robot histories feed the world stream. World and action streams share attention and use separate FFNs. Training-only heads (dashed) predict sparse whole-scene 3D displacement and visibility from patch states, and target SE(3) trajectories from the mean summary. Actions read scene and summary tokens through the structured mask. At inference, scene and summary keys and values are computed once per query and reused across 5 Euler steps; forecast heads are skipped. Geometry drawings are schematic.

3. SE(3)-WAM

We formulate dynamic manipulation as generating time-indexed command chunks (Sec. 3.1) from a representation supervised, at the execution times of these commands, with the scene's sparse 3D flow and the target's SE(3) trajectory (Fig. 2). We describe the metric inputs (Sec. 3.2), this supervision (Sec. 3.3), and how an action stream reads the representation to generate the chunk (Sec. 3.4).

3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query t, a policy π learned from demonstrations reads an RGB image I_t and a depth map D_t captured at time τ_t, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of H commands

$$(\mathbf{a}_{t,0}, \ldots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command j is meant for time $\tau_t + j\Delta_a$, with control period Δ_a.

Each task specifies a *target*, whose pose the robot must match within a contact or containment tolerance (Table S3). The surroundings that matter form the *context*: obstacles, which the robot must clear, and drivers, which move the target. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at τ_t.

3.2. Metric Scene and Motion Inputs

SE(3)-WAM encodes the target's motion precisely and its context through the current geometry: the current RGB-D frame gives the metric layout of the scene, and a short history of target poses gives the motion that one frame does not show, which also reflects how drivers have moved the target.

Current scene. SE(3)-WAM anchors each patch token at a metric 3D point, so that the observation shares the base frame of the forecasts and the actions. A frozen MAE ViT-B/16 [15] turns the RGB image into 196 patch tokens on a 14×14 grid, and the depth map back-projects the center of patch i to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

Target motion. Instead of past frames, from which motion must be recovered by comparing images, SE(3)-WAM receives the target's recent motion as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions

between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units (Appendix A.2). The velocity and acceleration that decide where the target will be thus enter the model directly, at the cost of a few low-dimensional tokens; Sec. 4.3 removes this history to measure its effect. In simulation, the poses come from the simulator; on the robot, a causal tracker estimates them from the RGB-D stream (Appendix D).

World stream. The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream.

3.3. Role-Aware 4D Predictive Supervision

SE(3)-WAM supervises its representation with two geometric forecasts of different precision: the motion of the whole scene as a sparse 3D flow on the patch grid, and, because the target's pose decides contact, the full SE(3) trajectory of the target. Both are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50\,\text{ms}$ for $h = 1, \dots, 8$, from 50 ms to 400 ms after capture; together with the current state, these horizons cover the execution times of all commands in the chunk (Appendix B.3).

Scene flow. At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for (Sec. 3.1). For each of the 196 points $\mathbf{x}_i$ and each horizon h, a linear head on the final state of patch token i predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where F_{ih} is the displacement of the surface point at $\mathbf{x}_i$, o_{ih} its visibility, s a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

Target trajectory. The flow samples the target only at the few patches it covers, too sparse to determine its rotation, so a dedicated linear head on the mean of the final summary states forecasts its rigid motion directly: for each horizon h, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\boldsymbol{\phi}}_h$, trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h\in\mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{3}$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t{+}\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^{\wedge})R(\tau_t)$, $d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. The rotation term matters for contact: a rotation error θ can displace a contact point at distance r from the target's origin by up to $2r\sin(\theta/2)$. For example, a 5° error at $r = 10\,\text{cm}$ can shift the contact point by nearly 9 mm.

3.4. World–Action Learning and Inference

Action stream. The action stream generates the chunk by conditional flow matching [22]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation increment, a rotation increment, and a gripper command (Appendix A.2). In every layer, one attention spans both streams with shared projections, followed by separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time. With S, M, and A denoting the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$, and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in X attend to tokens in $\mathcal{Y}$. The world stream never reads the actions. The action stream reads, in every layer, both the patch states shaped by scene-flow supervision and the summary states shaped by target-trajectory supervision.

Training objective. For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\ \mathbf{a}_1 - \mathbf{a}_0\big), \tag{4}$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_{\mathrm{aux}} \mathcal{L}_{\mathrm{aux}}, \tag{5}$$

where $\mathcal{L}_F$ and $\mathcal{L}_P$ are the flow and pose losses of Eqs. (2) and (3), and $\mathcal{L}_{\mathrm{aux}}$ sums auxiliary appearance losses (Appendix A.2).

Inference. At deployment, SE(3)-WAM skips the forecast heads, as recent world–action models do [20, 48, 51]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. Commands are indexed by time from capture, so the policy needs no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts

from the command indexed to that boundary (Appendix B.3). On the robot, the latency of a query also includes tracking the target in the newest frame; Sec. 4.2 reports the measured latency.

4. Experiments

We test whether role-aware precision and low latency together improve manipulation when the scene keeps moving during inference. We compare SE(3)-WAM with policies that range from reactive policies and VLAs acting on the current observation, through dynamic policies built for moving targets, to world–action models that forecast the whole scene, each under its own latency (Sec. 4.2). We then analyze which parts of its design account for the results (Sec. 4.3) and verify it on a real robot with tracked target poses (Sec. 4.4).

4.1. Experimental Setup

Dynamic-14. Unlike the evaluators of DOMINO [13] and PhysMani-Bench [49], which advance the simulation only after a policy has returned its prediction, Dynamic-14 shares one physical clock among observation, scene motion, and action execution, so the scene keeps moving while a policy computes and every method pays for its own latency. It contains fourteen tasks in MuJoCo/robosuite [37, 57] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Fig. 3; Appendix B). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all Dynamic-14 results on 1,400 held-out scenes (100 per task).

Evaluation. During each query, physics advances by the method's own inference latency, drawn from its latency distribution measured at batch size 1 on one RTX 5090, the GPU of our robot (Appendix B.3). All methods share the executor of Sec. 3.4, which starts each chunk at the command indexed to the control-period boundary at which the chunk arrives (Appendix B.3). Besides this native setting, we add 150 ms of latency to every method and, as a reference without the cost of computation, pause physics during inference. We report the task-balanced success rate (SR) and the p95 time-to-action, the measured time from an observation to its first command; paired 95% intervals come from a bootstrap over held-out scenes (Appendix B.4).

Baselines. We compare with reactive policies, dynamic VLAs, image- or latent-space WAMs, and a dynamic WAM (Table 1). Each baseline is trained on the same demonstrations with its official architecture and a task-adapted recipe, and its latency is measured in its own native deployment setting (Appendix B.5). SE(3)-WAM uses one agent view with its depth and the target's SE(3) history, without past frames. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

Table 1. **Main results.** Task-balanced SR on 1,400 held-out Dynamic-14 scenes: native latency, +150 ms, and paused physics. Input columns describe rollouts; p95 and replay budgets use each method's native deployment timing on one RTX 5090 (Sec. 4.1). OpenWAM uses official cache/compile; "off" disables both. Blue/bold and light blue rank available values; pending results are excluded.

Method	History			SR (%) ↑			
	Wrist RGB	Flow	SE(3)	Native	+150 ms	Paused	p95 (ms) ↓
ACT [54]				41.6	24.3	40.6	**5.8**
DP [11]	✓			30.7	15.9	31.0	68.9
$\pi_{0.5}$ [5]				**53.8**	27.8	66.0	59.6
$\pi_{0.5}$+SE(3)			✓	52.8	**28.2**	**74.9**	61.2
DynamicVLA [42]	✓			50.9	23.4	54.6	97.3
PUMA [13]		✓		—	—	—	—
OpenWAM-α [38]				46.1	24.7	58.3	151.7
OpenWAM-α (off)				15.8	10.4	58.3	466.3
DynamicWAM [25]		✓		40.5	20.6	49.1	149.6
DynamicWAM+SE(3)		✓	✓	36.8	21.9	64.1	153.3
SE(3)-WAM			✓	—	—	—	—

Implementation. The transformer has 24 layers of width 1024 with 16 heads and 523.8M trainable parameters; the MAE ViT-B/16 encoder and the DINOv3 teacher stay frozen. On Dynamic-14, it trains from random initialization with AdamW and $\lambda_F = \lambda_P = \lambda_{\text{aux}} = \lambda_R = 0.1$ for at most 20 epochs, and we keep the checkpoint with the lowest validation loss (Appendix A.3). On the real robot, it is fine-tuned per task from a 48-layer base pretrained on DROID [18] (Appendix A.4).

4.2. Dynamic Manipulation Results

Comparison. With its native latency, SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name], and [X.X] points above the strongest baseline with the same SE(3) history, [name]. Because the +SE(3) baselines receive the same target-motion input, the second margin measures what the representation of SE(3)-WAM adds beyond that input. Its p95 latency of [XX] ms is [X]× lower than that of the image-space WAMs and close to that of [the fastest reactive policy, to be fixed]. Its gains are largest on [task types] (Appendix C.1).

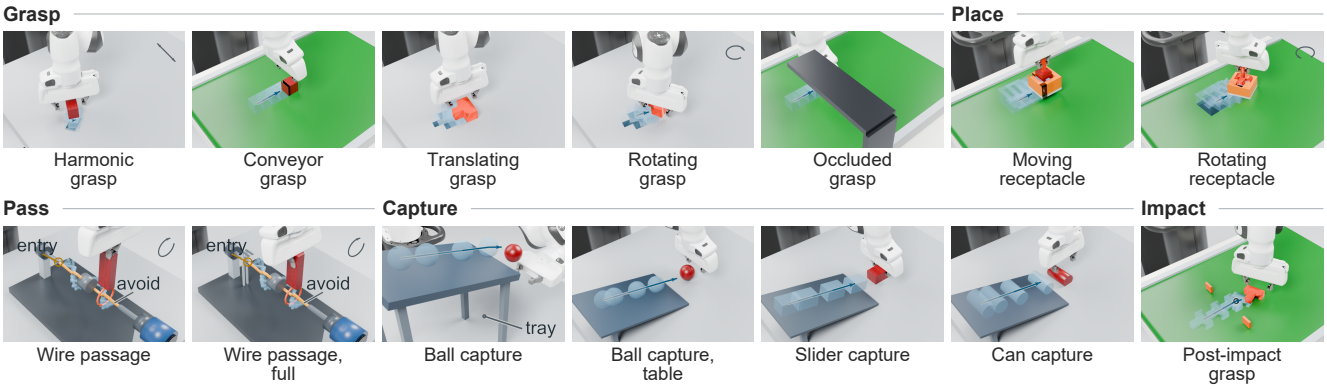


Figure 3. **Dynamic-14.** Fourteen tasks in five families, rendered from recorded simulator states with a common agent view. Four translucent blue poses fade from recent to earlier history; navy traces follow measured past motion. Gray corner arrows indicate rotation or oscillation qualitatively. A small ring marks the target center during recorded impact. Both Pass panels show passage: the wire is a target at entry and an obstacle along the path; Full additionally requires acquisition, exit and release. Occluded motion stays hidden. These native-expert examples illustrate task dynamics, not SE(3)-WAM outcomes; no future prediction is shown.

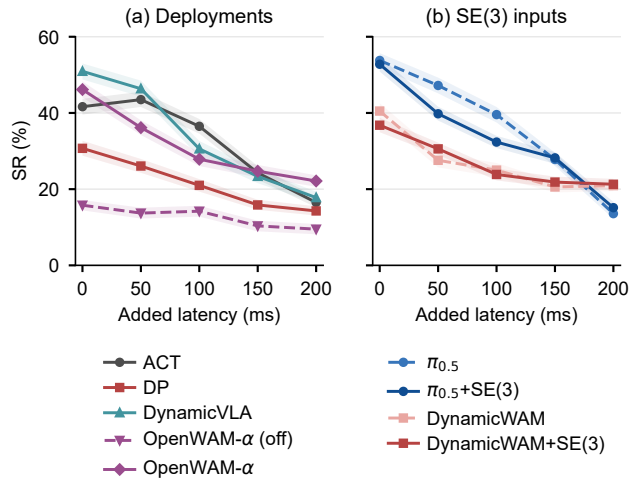


Figure 4. **Success under added latency.** (a) Baseline deployment configurations; OpenWAM uses official cache/compile, with both disabled in the off variant. (b) Paired comparisons with and without SE(3) inputs. Each point uses 1,400 held-out Dynamic-14 scenes and one training seed. Shading shows 95% task-stratified scene-bootstrap intervals, not training-seed variation. PUMA and final SE(3)-WAM results remain pending; all values: Table S7.

Effect of inference delay. [Charging computation changes the ranking:] with physics paused during inference, SE(3)-WAM leads [name] by [X.X] points, and with native latency by [X.X] points, while the image-space WAMs fall from [XX.X]% to [XX.X]% (Table 1). Adding 150 ms lengthens how far ahead every method must anticipate. SE(3)-WAM keeps [XX.X]% SR, whereas the reactive policies drop to [XX.X]% and the image-space WAMs to [XX.X]% on average: a method that does not anticipate the target's motion accumulates its velocity error over the longer delay. Fig. 4 shows the measured success rates from 0 to 200 ms of added latency.

Table 2. **DOMINO results** on 24 single-arm clean-L1 tasks, each method trained on DOMINO's demonstrations (50 episodes per task and method). SR (%) is grouped by interaction type and averaged over tasks; MS is the manipulation score. Best per column will be bold on blue once measured. Protocol: Appendix C.3.

Method	Intercept	Track	All ↑	MS ↑
$\pi_{0.5}$	—	—	—	—
DynamicWAM	—	—	—	—
PUMA	—	—	—	—
SE(3)-WAM	—	—	—	—

DOMINO. To test the design beyond Dynamic-14, we train SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [13] with its demonstrations and evaluate them with its native evaluator (Sec. 4.1). SE(3)-WAM reaches [XX.X]% SR against [XX.X]% for [the strongest baseline] (Table 2; Appendix C.3). As on Dynamic-14, it reads the simulator pose of the target.

4.3. Analysis of Role-Aware Modeling

Five controls each change one part of SE(3)-WAM and keep its data and recipe (Fig. 5; Appendix B.5). Except for the control without target history, each changes only training losses whose heads are skipped at deployment, so it shares the inference of SE(3)-WAM. We evaluate every control with native latency and with 150 ms of added latency.

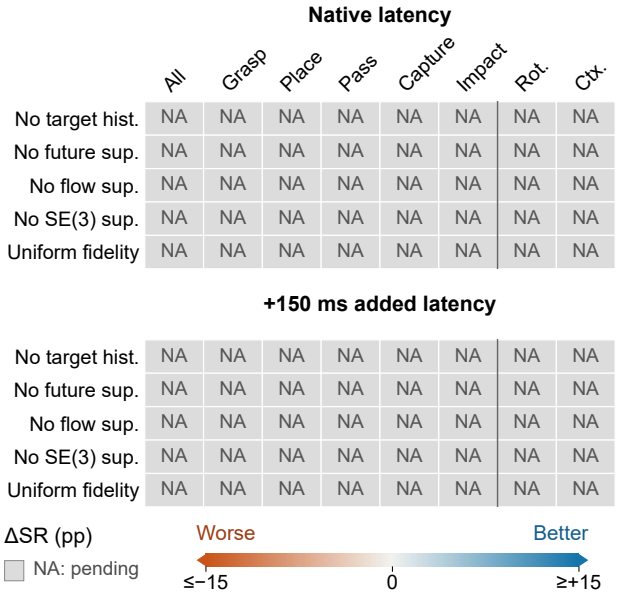


Figure 5. **Ablation effects by task (design preview; data pending).** ΔSR = SR(control) − SR(full), in percentage points; colors saturate at ± 15. Rows: the five controls of Sec. 4.3, in the order of the text; *uniform coarse flow* removes the dedicated SE(3) supervision, so that the target receives only the scene-flow supervision of every part of the scene. The full model reaches [XX.X]% SR with native latency and [XX.X]% with +150 ms. Rot./Ctx.: predeclared, overlapping rotation-sensitive/context-active groups (Table S3). Tasks are weighted equally, retaining `t3_phase`. Bold will mark unadjusted paired 95% scene-bootstrap intervals excluding zero, conditional on trained weights (Appendix B.4). Grey NA: no eligible paired results or estimates.

Table 3. **Real-robot results.** Success rate (%) on the six tasks of Sec. 4.4, 20 trials per task and method; All pools the trials. p95 lat.: p95 latency, the end-to-end time-to-action (ms) on one RTX 5090, including tracking for SE(3)-WAM. DP: Diffusion Policy. Once measured, best per column will be bold on blue, including ties. Successes, intervals, and failures in Appendix D.

Method	T1	T2	T3	T4	T5	T6	All ↑	p95 (ms) ↓
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
DP	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	—

Allocation of precision. The full model and both geometric ablations retain all auxiliary objectives. Without the dedicated SE(3) supervision, the target receives only the uniform coarse scene-flow supervision that every part of the scene receives, and SR drops by [X.X] points with native latency and by [X.X] points with 150 ms added. Without the scene-flow supervision, the target keeps its SE(3) trajectory while the rest of the scene, including the context, loses its motion supervision, and SR drops by [X.X] and [X.X] points. If precision follows the roles, the SE(3) supervision should matter most on the predeclared rotation-sensitive tasks, and the flow supervision on the context-active tasks, whose context constrains the robot or moves the target (Fig. 5; Appendix B.1). [Whether Fig. 5 shows this pattern, and what it implies for the allocation of precision.]

Target motion history. Removing the past target poses with their velocity and acceleration features, while keeping the current image, the current target pose, and the robot history, lowers SR by [X.X] points with native latency and by [X.X] points with 150 ms added. [The drop grows with the added latency,] as expected if the history supplies the target's velocity: a longer delay lets a velocity error accumulate further. Because the history relies on pose estimates, we also perturb them; under synthetic correlated pose noise, SE(3)-WAM keeps [XX.X]% SR (Appendix C.5).

Future supervision and auxiliary objectives. Removing all future supervision, that is, the SE(3), scene-flow, and future appearance targets, while keeping the reconstruction of the current target, leaves inference unchanged yet lowers SR by [X.X] points with native latency and by [X.X] points with 150 ms added, most on [tasks with non-uniform target motion]. Removing instead the auxiliary objectives $\mathcal{L}_{\text{aux}}$, including the current-target reconstruction, and keeping both geometric forecasts changes SR by [X.X] points, [what this implies for the role of the auxiliary objectives]. Offline, the learned SE(3) forecast is also more accurate than constant-velocity and constant-acceleration extrapolation of the same history (Table S9). [One-sentence summary of which components matter, and where.]

4.4. Real-Robot Evaluation

Setup. We evaluate six tasks on a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera (Fig. 6; Appendix D): conveyor grasping (T1), placement into a moving receptacle (T2), turntable grasping (T3), moving toy-car grasping (T4), tennis-ball grasping (T5), and block grasping from a ramp cart (T6). Baselines are $\pi_{0.5}$ (VLA), DynamicWAM (dynamic WAM), and Diffusion Policy [11] (reactive). All methods train or fine-tune on the same 100 teleoperated demonstrations per task and run 20 trials per task with shared trial keys (Appendix D.3). Baselines receive RGB [and a wrist camera if their recipe uses one]; SE(3)-WAM also receives FoundationPose [39] target estimates, updated at every control-period boundary before its query (Appendix D.2). End-to-end latencies include tracking (Table 3).

Results. SE(3)-WAM succeeds in [XX.X]% of all trials, against [XX.X]% for [the strongest baseline], a paired differ-

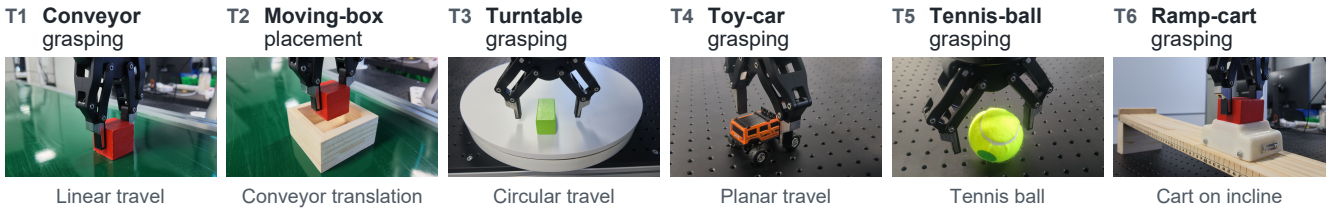


Figure 6. **Real-robot task setups.** Photographs of conveyor grasping (T1), placement into a moving box (T2), turntable grasping (T3), moving toy-car grasping (T4), tennis-ball grasping (T5), and grasping a block carried by a ramp cart (T6).

ence of [X.X] points (95% interval [[X.X, X.X]]; Table 3). Its p95 latency, including tracking, is [XX] ms, [within one control period], whereas [name] needs [XX] ms. The margin is largest on [tasks]. [One trajectory-verified contact example: how the target's pose, rotation, or context explains a success of SE(3)-WAM or a failure of a baseline.] The dominant failure of SE(3)-WAM is [failure mode] (Appendix D.4).

5. Conclusion

SE(3)-WAM spends precision where contact depends on it: a metric SE(3) history and forecast of the target and a sparse 3D flow of the whole scene, with the forecasts used only as training supervision. It reaches [XX.X]% success on Dynamic-14, [XX.X]% on DOMINO, and [XX]% on six real-robot tasks at a p95 latency of [XX] ms, improving on reactive, dynamic, and world–action policies. [The ablations show that the target history matters most under added latency and that future supervision improves control without changing the inference computation.] [State what the dedicated SE(3), scene-flow, and uniform-fidelity controls support: a better forecast design, or, if the task groups also agree, precision allocated by role.] Accounting for inference latency [state how success rates and relative performance differ from paused evaluation].

Limitations. SE(3)-WAM requires a specified rigid target and reliable estimates of its pose, and its forecast does not model how the robot's own contact changes the target's motion. Its target-trajectory head predicts a single future per horizon and does not represent uncertainty or multimodal motion, which an uncertainty-aware forecast could address.

LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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